

0.1 Randomization test for the R -threshold analysis

To assess whether the observed relationship between solar-wind density and earthquake occurrence could arise by chance, we performed a Monte Carlo randomization test based on the event relative rate, R , as a function of the solar-wind-density threshold. The analysis was performed using daily OMNI solar-wind data and the ISC-GEM earthquake catalog. Only days for which valid OMNI data were available were included in the analysis. Earthquakes were first filtered using a minimum moment-magnitude threshold, $M_w \geq 5.6$, and then mapped onto the valid OMNI days to create a daily earthquake-count time series.

For a given solar-wind variable V , in this case the proton density, we define a set of normalized thresholds,

$$V_{\text{step}} = \frac{V_T - V_{\min}}{V_{\max} - V_{\min}}, \quad (1)$$

where V_T is the physical threshold value, and $V_{\min}$ and $V_{\max}$ are the minimum and maximum values of the solar-wind variable over the analyzed interval. For each threshold, we identify the set of days satisfying a given threshold condition. Examples include days above the threshold, days below the threshold, the last day above the threshold before a drop, and the first day below the threshold after a drop.

For each threshold condition, the event relative rate is defined as

$$R = \frac{E_C/D_C}{(E - E_C)/(D - D_C)}, \quad (2)$$

where D is the total number of valid OMNI days, E is the total number of earthquakes occurring on those valid days, D_C is the number of days satisfying the selected threshold condition, and E_C is the number of earthquakes occurring on those condition days. Thus, E_C/D_C is the earthquake occurrence rate during the selected condition, while $(E - E_C)/(D - D_C)$ is the earthquake occurrence rate outside the selected condition. Therefore, $R > 1$ indicates that earthquakes occurred more frequently during the selected condition than outside it, while $R < 1$ indicates a lower occurrence rate during the selected condition. If earthquake occurrence is unrelated to the solar-wind threshold condition, then R should fluctuate around unity, apart from random statistical variability.

The real R curve was first computed using the observed earthquake daily-count series and the observed OMNI solar-wind-density record. We then generated surrogate earthquake catalogs using a circular-shift randomization procedure. In each Monte Carlo realization, the daily earthquake-count series was shifted by a randomly selected number of days relative to the fixed OMNI time series:

$$N_{\text{eq}}^{(i)}(t) = N_{\text{eq}}(t + \Delta t_i), \quad (3)$$

where Δt_i is a random circular time shift for the i th realization. This procedure preserves the internal temporal structure of the earthquake daily-count series, including clustering and periods of enhanced or reduced activity, but destroys the specific timing alignment between earthquake occurrence and the solar-wind-density record. The OMNI data are not shuffled in this test, so the temporal structure of the solar-wind density, including threshold crossings and multi-day enhancements, is preserved.

For each randomized earthquake catalog, the full R -threshold curve was recomputed using exactly the same thresholds and threshold conditions as were used for the real data. This produced an ensemble of randomized curves,

$$R_{\text{rand},i}(V_T), \quad i = 1, 2, \dots, N_{\text{rand}}, \quad (4)$$

which represents the distribution of R values expected if the timing of earthquakes were unrelated to the solar-wind-density threshold condition.

We used two complementary measures to interpret the results. First, we computed pointwise confidence envelopes for R at each threshold. At each value of V_T , the real value $R_{\text{real}}(V_T)$ was compared with the distribution of randomized values $R_{\text{rand}}(V_T)$. If the real curve lies within the random envelope, then the observed value at that threshold is consistent with chance alignment. If the real curve exceeds the upper tail of the randomized distribution, then the earthquake occurrence rate for that condition is unusually high compared with the circular-shift null model.

Second, because many thresholds are tested, we also computed a global maximum- R statistic. For the real data, this statistic is

$$R_{\text{max,real}} = \max_{V_T} R_{\text{real}}(V_T). \quad (5)$$

For each randomized realization, we similarly compute

$$R_{\max,\text{rand},i} = \max_{V_T} R_{\text{rand},i}(V_T). \quad (6)$$

The global Monte Carlo p -value is then estimated as

$$p_{\text{global}} = \frac{\#(R_{\max,\text{rand},i} \geq R_{\max,\text{real}})}{N_{\text{rand}}}. \quad (7)$$

This value gives the fraction of randomized catalogs that produce a maximum R value at least as large as the real maximum R anywhere in the scanned threshold range. This is a more conservative test than a pointwise comparison, because it accounts for the fact that scanning many thresholds can produce an apparently large peak by chance.

The quantity reported in the code as `p_global_max_R` should therefore be interpreted as the probability, under the circular-shift null model, of obtaining an R peak as large as the observed one somewhere in the tested threshold range. A large value of `p_global_max_R`, for example $p_{\text{global}}0.1$, indicates that the observed maximum R is commonly reproduced by randomized earthquake catalogs and is therefore consistent with random temporal alignment. A small value, for example $p_{\text{global}} < 0.05$, indicates that the observed peak is relatively unusual under the null model. If no randomized realization produces a maximum R as large as the real one, the result should not be reported as exactly $p = 0$; instead, for N_{rand} randomizations, it should be reported as $p_{\text{global}} < 1/N_{\text{rand}}$, or conservatively as $p_{\text{global}} \approx 1/(N_{\text{rand}} + 1)$.

In summary, this randomization test asks whether the observed R -threshold pattern is stronger than expected from random timing between earthquakes and solar-wind density, while preserving the temporal structure of both datasets as much as possible. If the real R curve and its maximum value are comparable to those obtained from the circularly shifted catalogs, then the apparent threshold-dependent enhancement in earthquake occurrence is statistically indistinguishable from random alignment. Conversely, if the real curve exceeds the randomized envelope and has a small global maximum- R probability, then the observed enhancement is unlikely under this particular null model.